TEMPORARY EPICARDIAL PACING

PROCEDURE _____ OP DATE _____ DAY ___

GENERATOR / BOX _____ SET BY _____

WIRE MAP & PORTS

RA 1 | RA 2 white · R of sternum → A ports

RV 1 | RV 2 blue · L of sternum → V ports

Correct A/V ports · polarity within a pair irrelevant

PATIENT / ADDRESSOGRAPH

① FROM POWER-ON — INITIALISATION

top → bottom

1. **PREPARE** — non-conductive gloves; discharge static; check **battery**. Identify A white/R, V blue/L.
2. **CONNECT** to correct **A & V ports**; secure. Polarity within a pair irrelevant.
3. **POWER ON & UNLOCK** — box must be ON to program (nominal DDD 60).
4. **SELECT MODE** for the rhythm — see Mode Select →.
5. **SET RATE** — 80–90 bpm post-op to optimise CO (or as indicated).
6. **SET OUTPUT HIGH** — start ~10 V to guarantee capture before titrating.
7. **CONFIRM CAPTURE** — electrical (spike → P/QRS) **and** mechanical (art / SpO_2).
8. **MEASURE THRESHOLDS** — capture, then sensing (→). Output **≥5 V or 3× thr**; sensitivity **≈½ thr**.
9. **CONFIRM SENSING** — LED tracks each intrinsic P/R; no competing spikes / inhibition.
10. **LOCK · ALARM · DOCUMENT** — key-lock; rate/alarm limits; baseline recorded; defib + transcutaneous ready.

CAPTURE (STIMULATION) THRESHOLD

1. VVI (atrium: AAI); rate **+10 bpm** over intrinsic so box paces 1:1.
2. From a capturing output, **decrease slowly** until capture lost — spike, no P/QRS, loss of art/SpO_2.
3. **Increase** until capture returns = **capture threshold** (acute V usually <1 V).
4. Set output **≥5 V or 3× threshold**.

SENSING THRESHOLD (SENSITIVITY)

1. Demand mode; rate **below intrinsic** (beats visible, box inhibited).
2. **Raise the number** (↓ sensitivity) until sense LED stops / reverts to asynchronous — **do not sustain** (R-on-T).
3. **Lower the number** until LED tracks each beat = **sensing threshold**.
4. Set **≈ ½ threshold**. No rhythm → **2 mV**. (adequate: R>5, P>1.5–2 mV)

Reminder: **LOWER number = MORE sensitive.**

MODE SELECT

rhythm → mode

Situation	Mode
Sinus brady, intact AV, atrium capturable	**AAI**
AF / atrium uncapturable, need rate	**VVI**
AV block, sinus present, want synchrony	**DDD**
AV block + atrial tachyarrhythmia	**DDI** / mode-switch
AV synchrony + diathermy / PMT	**DOO**
Standard post-op, optimise CO	**DDD** 80–90
No rhythm / crash	**VVI/DOO**, max o/p

⚠ EMERGENCY / CRASH PACE

1. Mode **VVI** (or **DOO** if interference) · Rate **80** · Output **MAX (10 V)** · Sensitivity nominal.
2. Confirm **MECHANICAL** capture (art/SpO_2), not ECG alone.
3. No capture → **transcutaneous pace + defib**; treat K^+, acidosis, hypoxia.

DAILY CHECK PROTOCOL — in this order, every shift & after any change

stop at any failed step and fix it first

1. **SAFETY** — gloves, discharge static, insulated wire ends, defib + transcutaneous pads in reach. ECG monitor in **pacing / spike-enhanced** mode.
2. **GENERATOR** — ON, unlocked, battery OK, connections seated, A/V ports correct. Record current **mode / rate / outputs / sensitivities**.
3. **CONFIRM CAPTURE NOW** — each spike → P/QRS **and** a pulse on art / SpO_2. If not → Loss-of-Capture ladder (p4) *before* any threshold test.
4. **UNDERLYING RHYTHM (pause test)** — briefly drop rate below intrinsic (or ≈40). Note rhythm & rate. **No escape rhythm → STOP: do not test thresholds.**
5. **CAPTURE THRESHOLD** — VVI/AAI +10 bpm; ↓ output to loss, ↑ to return. Record A & V.
6. **SET OUTPUT** — ≥5 V or 3× threshold (greater). Re-confirm mechanical capture.
7. **SENSING THRESHOLD** — rate below intrinsic; ↑ number until LED stops, ↓ until it tracks every beat. Record A & V.
8. **SET SENSITIVITY** — ≈ ½ threshold (no rhythm → 2 mV). Watch 30 s: no competing spikes, no pauses.
9. **RESTORE** intended mode & rate · **LOCK** · set alarm limits.
10. **DOCUMENT & TREND** — flowsheet (p6). Rising threshold vs yesterday → anticipate failure; day 4–5 → consider PPM.

NBG: I paced (O/A/V/D) · II sensed (O/A/V/D) · III response (I inhibited · T triggered · D dual · O none). **Defaults:** start-up DDD 60 → theatre exit DDD 80–90 · chronic AF → VVI/DVI.

Ref: Southampton Cardiac Anaesthesia — epicardial pacing; standard CICU practice. Reference aid — follow local protocol, device IFU and senior guidance. Schematic traces not to scale.

TEMPORARY EPICARDIAL PACING

PACED-ECG RECOGNITION

red line = pacing artefact (spike)

ATRIAL PACED (AAI)



Spike → **P wave** → conducted **narrow** QRS. Atrial capture with intact AV conduction.

VENTRICULAR PACED (VVI)



Spike → **wide** QRS (LBBB-like) with **discordant** T. No preceding P.

AV SEQUENTIAL (DDD/DVI)



Two spikes: atrial → P, then ventricular → wide QRS. Restores AV synchrony / atrial kick.

FUSION



Spike coincides with intrinsic depolarisation → **hybrid** QRS. Normal, not a fault.

PSEUDOFUSION



Spike falls on intrinsic QRS, **morphology unchanged**. Normal; do not mistake for failure to capture.

FAILURE TO CAPTURE



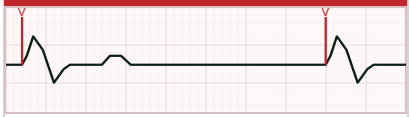
Spike with **no following** P/QRS. Confirm true (spike-enhance), ↑ output, check reversibles/circuit.

UNDERSENSING



Spike fires despite intrinsic beat → **competing** pacing, R-on-T risk. ↑ sensitivity (LOWER number).

OVERSENSING



Inappropriate **inhibition** → unexpected pause (e.g. T-wave / diathermy sensed). ↓ sensitivity (RAISE number).

ADVANCED TECHNIQUES

ATRIAL ELECTROGRAM (AEG) — DIAGNOSE THE RHYTHM

1. Use the **atrial epicardial wires as recording electrodes** to magnify atrial activity.
2. **Unipolar**: one atrial wire → a chest (V) lead. **Bipolar**: two atrial wires → RA & LA limb leads, read **lead I**.
3. Reveals P–QRS relationship → **SVT vs VT**, flutter, AV dissociation, diagnoses broad-complex tachycardia.
4. Label the trace; keep skin/leads insulated (micro-shock).

ATRIAL OVERDRIVE / BURST PACING — FLUTTER

1. Type I atrial flutter / re-entrant SVT with atrial wires that **capture**.
2. AAI at **10–20% above** the flutter rate, or **burst** ~400–600/min for a few seconds, then abruptly stop.
3. Aim: capture the circuit & interrupt re-entry. **Risk: induces AF** — have defib ready; not for AF.

AV DELAY · CROSS-TALK · SAFETY PACING

1. **AV delay** (DDD) ~120–200 ms: shorter → more V pacing; longer → favours intrinsic conduction. Optimise for filling/CO.
2. **Cross-talk**: atrial output oversensed by the V channel → inappropriate V inhibition. Guarded by blanking/refractory periods.
3. **Ventricular safety pacing**: committed V pace at short AV (~110 ms) if a signal is sensed early — prevents asystole from cross-talk.

PMT · RUNAWAY · WEANING

1. **PMT** (endless-loop, tracking modes via retrograde VA): terminate with **magnet / DOO**, switch to **DDI/VVI**, or ↑ PVARP.
2. **Runaway pacemaker** (rare generator fault, rapid pacing): **disconnect wires / replace box** — emergency.
3. **Weaning**: drop rate below intrinsic; confirm stable escape rhythm + haemodynamics; leave **back-up at 40**.

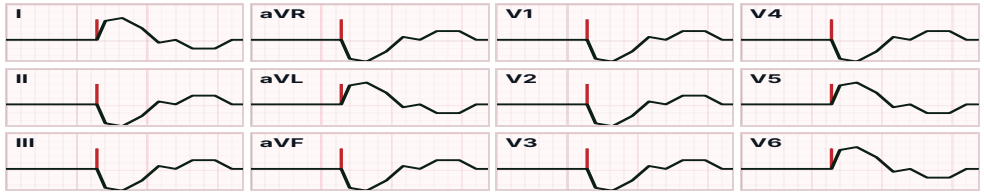
KEY NUMBERS

Capture	Sensing	Intervals	Overdrive
Acute V threshold <1 V (atrium higher). Output ≥5 V or 3× thr. Rises after ~day 4.	Set ≈½ threshold. Adequate signal R>5 mV, P>1.5–2 mV. No rhythm → 2 mV.	Rate 80–90 post-op · AV delay 120–200 ms · safety-pace AV ~110 ms · back-up 40.	Flutter: +10–20% or burst 400–600/min. Defib available; may induce AF.

Ref: Southampton Cardiac Anaesthesia — epicardial pacing; standard CICU practice. Reference aid — follow local protocol, device IFU and senior guidance. Schematic traces not to scale.

12-LEAD — RV-PACED RHYTHM (schematic)

red = pacing spike



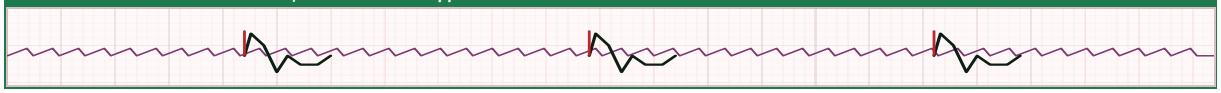
READING A PACED 12-LEAD

1. **RV apex** (usual temporary site) → **LBBB** morphology + **left/superior axis**: II, III, aVF **negative** (QS); I, aVL positive.
2. **Precordial**: LBBB pattern — V1–V3 dominantly **negative**.
3. **RVOT** pacing → LBBB + **inferior** axis (II/III/aVF **positive**, tall R).
4. **LV / epicardial** wires → **RBBB-like** (dominant R in V1).
5. Expect **appropriate discordance** (ST/T opposite to main QRS). Narrow or absent discordance → suspect **fusion / pseudofusion / lead displacement**.
6. Diagnosing ischaemia in a paced/LBBB rhythm → **Sgarbossa** criteria (concordant ST↑, or excessive discordant ST↑).

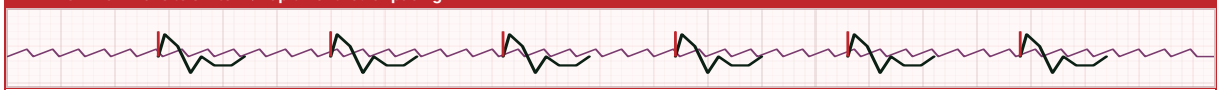
MODE SWITCH — DDD DURING ATRIAL TACHYARRHYTHMIA

appropriate vs failure modes

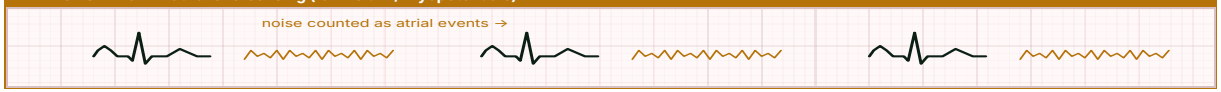
✓ APPROPRIATE — switch DDD→DDI, ventricular rate capped



× TRACKING — fails to switch → rapid ventricular pacing



× FALSE SWITCH — atrial oversensing (far-field R / myopotentials)



ON THE TEMPORARY BOX

1. Fast atrial rhythm tracked → **manually switch DDD → DDI or VVI**; consider ↓ upper-rate limit.
2. Then **treat the arrhythmia**: atrial overdrive/burst (flutter), rate control, or DC cardioversion.
3. **False switch** (loss of AV synchrony in true sinus) → **reduce atrial sensitivity** (raise number), check lead contact.

SITE → MORPHOLOGY

RV apex LBBB + superior (left) axis. II/III/aVF negative.	**RVOT** LBBB + inferior axis. II/III/aVF positive.	**LV / epicardial** RBBB-like; dominant R in V1.	**Ischaemia in paced rhythm** Apply Sgarbossa criteria (concordant / excessive discordant ST).

Ref: Southampton Cardiac Anaesthesia — epicardial pacing; standard CICU practice. Reference aid — follow local protocol, device IFU and senior guidance. Schematic traces not to scale.

LOSS OF CAPTURE — spike present but no myocardial response (electrical & / or mechanical)

work top → bottom, in parallel if compromised

⚠ TRAP — AT MAX SENSITIVITY & LOSING CAPTURE · SENSITIVITY IS NOT THE CAPTURE DIAL

› **Sensitivity = input (sensing); capture = output.** Turning sensitivity ↑ **cannot** restore capture.

› **At MAX sensitivity** you oversense (T-wave / myopotential / diathermy) → box **inhibits** output → **missing spikes**, easily misread as 'lost capture'.

› **Spike, no P/QRS** → **true** capture failure → work the **OUTPUT** ladder; sensitivity is irrelevant.

› **Spikes missing / pauses** → **oversensing** → **reduce** sensitivity (**RAISE** the number), not raise it.

› **If V-dependent → go VOO at MAX output.** Asynchronous = no sensing → removes over- *and* under-sensing at a stroke.

› **Captures in VOO** → it was a **sensing** problem, not capture; once stable reset sensitivity ≈ ½ threshold.

› **No capture in VOO at max output** → **true capture failure** → emergency: transcutaneous → transvenous → iso / adrenaline; treat reversibles; call for help.

› Already at max sensitivity ⇒ **functional** non-capture (undersensing) is **excluded** → it's **output / circuit / excitability**.

A · CONFIRM & PROTECT

1. **CONFIRM it is real** — monitor to **pacing / spike-enhanced** mode; spike with no P/QRS? Check **mechanical** (art / SpO_2 / pulse), not ECG alone.
2. **PROTECT** — if compromised or **no escape rhythm**: call for help, apply **transcutaneous pads**, defib to hand, support circulation.

▼

B · OUTPUT

3. **Increase output to MAX** (chamber-specific, A and/or V). Recaptures? → **threshold has risen** (commonest after a few days) — leave high, plan review.

▼

C · CIRCUIT / MECHANICAL

fast, often the cause

4. **Connections** — reseat wire→cable and cable→generator. Box **ON & UNLOCKED?**
5. **Battery** — insert / replace fresh.
6. **Wire** — inspect for **fracture / dislodgement** (frequent culprit).
7. **Polarity / switch** — bipolar→**unipolar** (other wire, or skin electrode as anode); try the **alternate wire pair**.

▼

D · REVERSIBLES then ESCALATE

8. **Correct reversible contributors** — see box →.
9. **Still failing & dependent** — hold max output + transcutaneous; start **isoprenaline / adrenaline** as a chemical bridge; arrange **transvenous** wire; call cardiology / cardiac surgery.
10. **Document & escalate** — thresholds, actions, plan. Consider **permanent system**.

REVERSIBLE CONTRIBUTORS (raise threshold / exit block)

1. **Electrolytes**: **hyperkalaemia** (key), hypokalaemia, hypomagnesaemia.
2. **Acid–base / gas**: acidosis, hypoxia, hypercapnia.
3. **Drugs**: class I (**flecainide**) & III antiarrhythmics; others.
4. **Metabolic**: hypothermia, hypothyroidism.
5. **Myocardial**: ischaemia / infarction; interface **oedema / fibrosis** (day 4+); perforation.

MIMICS — is it really loss of capture?

1. **No spike at all** → failure to **OUTPUT** (battery / lead / oversensing), not capture.
2. Spike in **refractory tissue** from **undersensing** → **functional** non-capture → fix sensing.
3. Spike on intrinsic QRS, unchanged complex → **pseudofusion** (normal).
4. Progressive ↑ output over days → **exit block** / fibrosis → plan PPM.

⚠ NO ESCAPE RHYTHM = PERI-ARREST

1. Persistent non-capture + no underlying rhythm → treat as arrest pathway.
2. **Transcutaneous pace now** → **transvenous** temporary wire; **iso / adrenaline** bridge.
3. Treat reversibles aggressively; urgent surgical / cardiology review; ?permanent system.

CAPTURE — REMEMBER

Chamber-specific Atrial capture is harder; atrial wires fail earlier (>60% R, >80% L by day 5).	**Electrical ≠ mechanical** Confirm capture on art / SpO_2, never the ECG spike alone.	**Thresholds drift up** Interface oedema / fibrosis from ~day 4 → output ≥5 V or 3× threshold.	**Undersensing masquerade** A spike in refractory tissue looks like non-capture — check sensing.

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SENSING FAULTS — the box sees too much (over) or too little (under). Direction rule: LOWER number = MORE sensitive. fix the side that matches the trace

OVERSENSING — inappropriate inhibition → pauses

fix = RAISE the number

1. **RECOGNISE** — unexpected **pauses** / missing spikes, rate below set rate, sense LED flashing *without* a P/QRS. Dependent patient → may be asystole.
2. **PROTECT** — if dependent or symptomatic: switch to **VOO / DOO** (asynchronous) *now*. Pads on, defib ready.
3. **IDENTIFY THE SIGNAL** — T-wave? far-field R (atrial channel)? myopotentials (shiver, movement)? **diathermy** / electrical interference? loose connection noise?
4. **↓ SENSITIVITY — RAISE the number** stepwise (e.g. 2 → 3 → 5 mV) until pauses stop *and* intrinsic beats are still sensed.
5. **REMOVE THE SOURCE** — stop shivering/movement, bipolar diathermy & short bursts, re-route cables, reseat connections, change ECG cable.
6. **STILL OVERSENSING** → try the **other wire pair** / bipolar config; leave **asynchronous** if dependent; senior review.
7. **RE-MEASURE & DOCUMENT** — repeat sensing threshold; reset ≈ ½; log change & cause.

UNDERSENSING — missed beats → competing spikes

fix = LOWER the number

1. **RECOGNISE** — spikes marching **through** intrinsic beats, **competing** paced & native QRS, spikes landing on T (**R-on-T**), sense LED *not* flashing with native beats.
2. **PROTECT** — R-on-T risk: defib ready. If native rhythm adequate, temporarily **↓ rate below intrinsic** so spikes stop competing while you fix.
3. **↑ SENSITIVITY — LOWER the number** stepwise (e.g. 5 → 3 → 2 → 1 mV) until LED tracks every native beat.
4. **SIGNAL TOO SMALL?** — re-measure threshold; adequate R >5 mV, P >1.5–2 mV. Small signal → poor contact / fibrosis / infarct / electrolytes.
5. **CHANGE THE CIRCUIT** — reseat connections; **switch wire pair**; bipolar ↔ unipolar (skin electrode as indifferent).
6. **CAN'T SENSE RELIABLY** → if native rhythm sufficient, pace **below** it (back-up ~40); if dependent, accept **VOO/DOO** & plan transvenous / PPM.
7. **RE-MEASURE & DOCUMENT** — threshold, setting, cause. Check for **functional non-capture** that the undersensing was causing.

WHEN TO GO ASYNCHRONOUS (VOO / DOO)

1. **Dependent** patient with oversensing pauses — immediately.
2. Diathermy / electrocautery in use.
3. Unable to find a sensitivity that both tracks native beats and avoids pauses.
4. **Never** leave asynchronous with a competing native rhythm unless unavoidable — R-on-T risk; defib ready.

ATRIAL-CHANNEL SPECIFICS (DDD)

1. Atrial **oversensing** of far-field R / myopotentials → **false mode-switch** or inappropriate inhibition → raise atrial number; check PVAB/blanking.
2. Atrial **undersensing** → box paces atrium into native P → **loss of AV synchrony**; tracking failures → lower atrial number.
3. Rapid atrial rhythm tracked at upper rate → switch **DDI/VVI**, then treat the arrhythmia.

CROSS-TALK & SAFETY PACING

1. **Cross-talk**: atrial stimulus sensed on the V channel → V output **inhibited** → ventricular asystole in CHB.
2. Protective: **ventricular blanking** after A-pace + **safety pacing** (committed V pace at ~110 ms) on some boxes.
3. If suspected: ↓ atrial output, ↑ V number slightly, or **DOO/DVI**; confirm V capture mechanically.

SENSING — REMEMBER

Number direction Lower mV = more sensitive. Oversense → raise. Undersense → lower.	**Set ≈ ½ threshold** Margin for small signals & fibrosis. Adequate R >5, P >1.5–2 mV.	**Max sensitivity is a trap** Lowest number → T-wave / noise oversensing → pauses that mimic lost capture.	**Asynchronous = no sensing** VOO/DOO cuts both faults at a stroke; use for dependence & diathermy, defib ready.

Ref: Southampton Cardiac Anaesthesia — epicardial pacing; standard CICU practice. Reference aid — follow local protocol, device IFU and senior guidance. Schematic traces not to scale.

TEMPORARY EPICARDIAL PACING

ROUTINE CHECKS

each shift & after every change

Parameter					
Date / time					
Battery / display OK ✓					
Mode set					
Rate set (bpm)					
Underlying rhythm — pause test					
Sensitivity — Atrial (mV)					
Sensitivity — Ventric. (mV)					
Capture threshold — A (V)					
Capture threshold — V (V)					
Output set — A (V)					
Output set — V (V)					
Mechanical capture ✓ (art/SpO_2)					
Wire site clean / dressed ✓					
Micro-shock precautions ✓					
Checked by (initials)					

IABP + PACING

1. Pacing spikes may be read as QRS by ECG-triggered IABP.
2. Keep **HF filter ON**, or trigger off the **arterial** waveform.

CONSIDER PERMANENT PPM

1. Risk: age, **pre-op BBB**, long CPB, poor myocardial protection.
2. Indications: CHB, SND, slow-response AF, Mobitz II. Threshold drift day 4–5.

WIRE REMOVAL

1. ~Day 4 on surgeon's instruction; steady gentle traction. If caught, cut at skin (ends retract).
2. Stop heparin → remove after **2 h** → drain out +1 h if dry → restart heparin.
3. Observe for **tamponade**, VT, anastomotic injury.

BACK-UP PACING

1. Rate ~40 lets intrinsic rhythm persist until compromise.
2. Enables continuous sensing-threshold monitoring.

TROUBLESHOOTING

work the failing element

① **FAILURE TO CAPTURE** — spike, no P/QRS — **full ladder p4**

▼ Confirm — monitor to **pacing / spike-enhanced**

▼ Increase **output to max** → recaptures? = rising threshold

▼ Reversible: **K^+**, acidosis, hypoxia, antiarrhythmics, hypothermia

▼ Circuit: connections · battery · fracture · **reverse polarity** / other pair

▼ **No rhythm = EMERGENCY** → max output · **transcutaneous** · defib

② **FAILURE TO SENSE** — **full ladder p5**

OVER	UNDER
pauses / inhibition	competing spikes · R-on-T
↓ **sensitivity = RAISE number**	↑ **sensitivity = LOWER number**
Remove T-wave / diathermy. Dependent → **VOO/DOO**	Signal small → check contact · **switch pair**

③ **FAILURE TO OUTPUT** — no spike

▼ Box **ON & UNLOCKED?** (on to program, then key-locked)

▼ Battery · loose / fractured wire · connection

▼ Oversensing inhibiting → treat as ② over-sense

TARGETS & SAFETY

Sensitivity
LOWER number = MORE sensitive.
Oversense→raise; undersense→lower. No rhythm→2 mV.

Output
≥5 V or 3× threshold. Confirm **MECHANICAL** capture on art/SpO_2.

No-rhythm rule
Never test thresholds without an escape rhythm — watch for dropped beats.

Micro-shock
Non-conductive gloves; discharge static; insulate wires. ECG + defib w/ transcutaneous always.

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